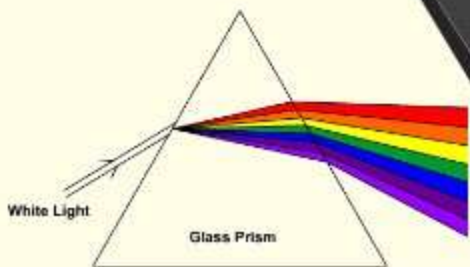


PHYSICS

SOUND WAVES

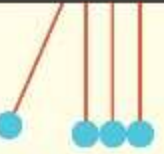
MODULE - 13



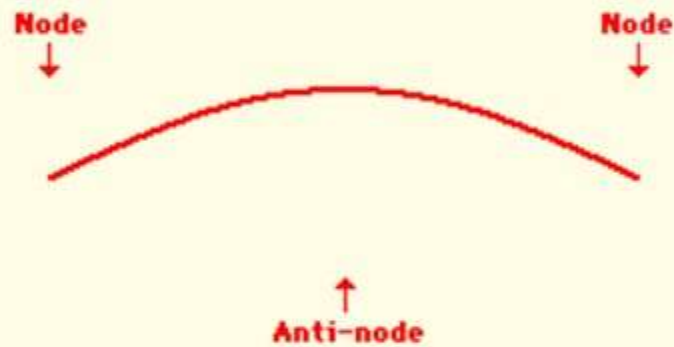
HOW IS MUSIC FORMED BY THE INSTRUMENTS?



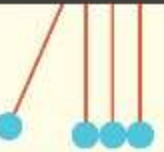
When we pluck a string or strike a chord or play through the flute, only certain frequencies of stationary waves are able to form, these are called standing waves



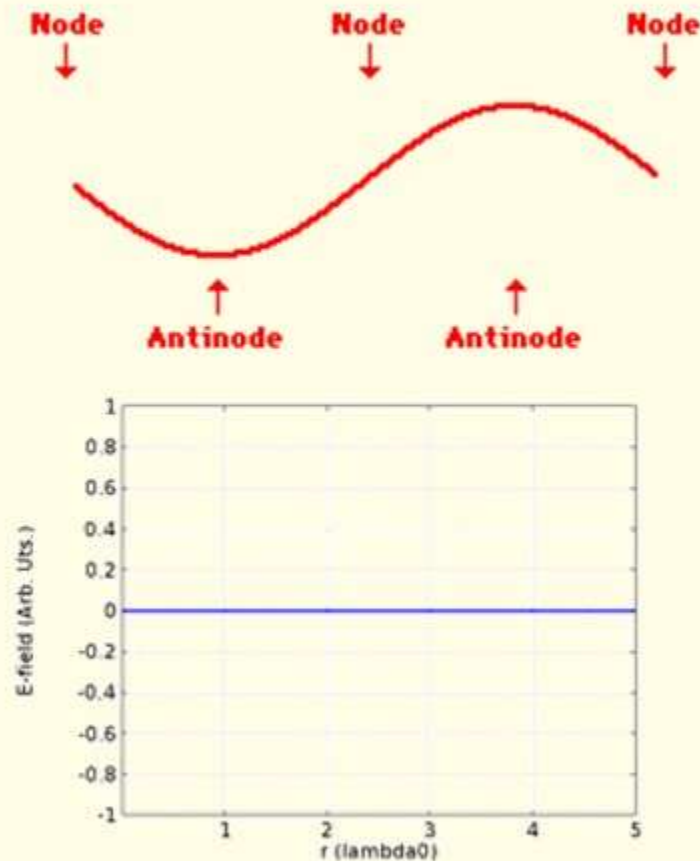
STANDING WAVES



- A vibration of a system in which some particular points remain fixed while others between them vibrate with the maximum amplitude.
- The simplest standing wave pattern that could be produced that has points of no displacement (nodes) at the two ends.
- It has only one point of maximum displacement (antinode) in the middle.
- The above standing wave pattern is known as the first harmonic.



STANDING WAVES

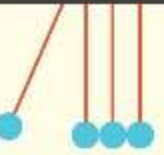


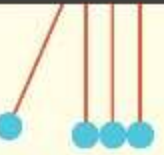
- This standing wave pattern is characterized by nodes on the two ends and an additional node in the exact center.
- As in all standing wave patterns, every node is separated by an antinode.
- This pattern with three nodes and two antinodes is referred to as the second harmonic

SOUND WAVES



A sound wave is a disturbance that is transported through a medium via the mechanism of particle-to-particle interaction. a sound wave is characterized as a mechanical wave.





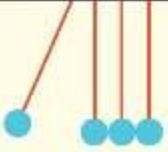
QUESTION 1

A sound wave is different than a light wave in that a sound wave is

- a. produced by an oscillating object and a light wave is not.
- b. not capable of traveling through a vacuum.
- c. not capable of diffracting and a light wave is.
- d. capable of existing with a variety of frequencies and a light wave has a single frequency.



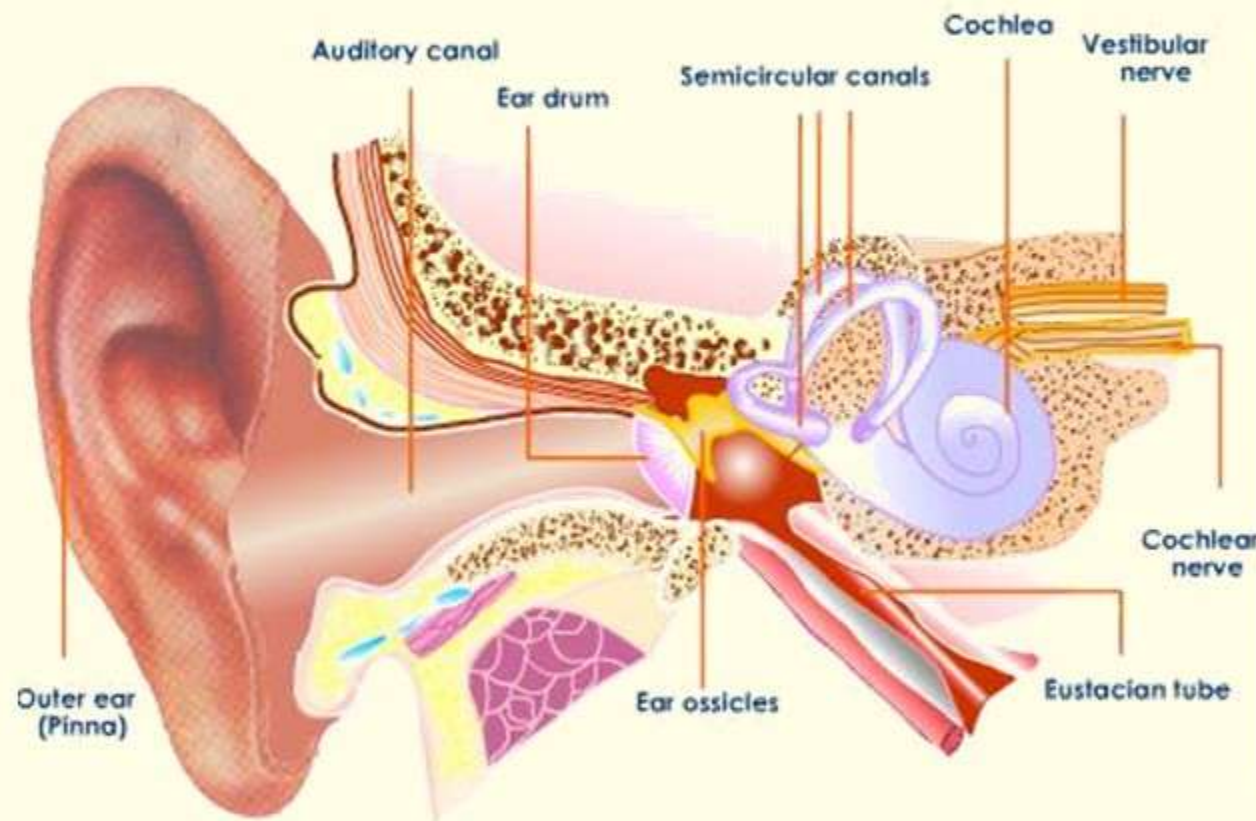
HOW SOUND WAVES ARE DETECTED IN OUR EARS?



- Since a sound wave consists of a repeating pattern of high-pressure and low-pressure regions (compression and rarefaction) moving through a medium,
- it is sometimes referred to as a pressure wave.
- If a detector, whether it is the human ear or a man-made instrument, were used to detect a sound wave,
- It would detect fluctuations in pressure as the sound wave impinges upon the detecting device.

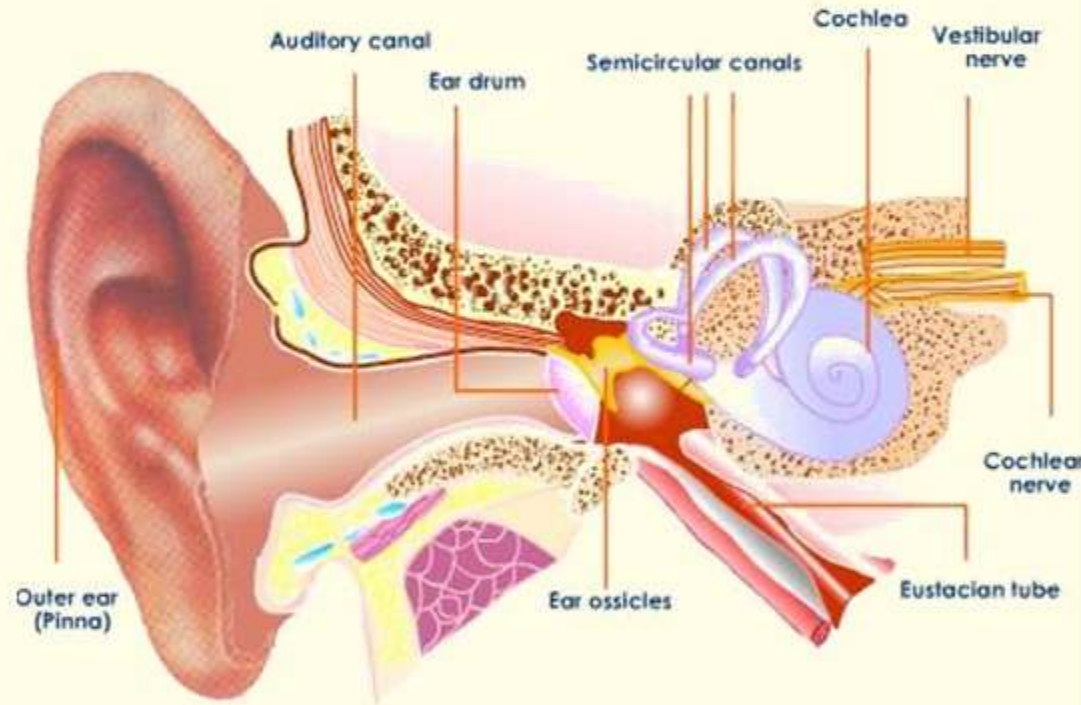


HOW SOUND WAVES ARE DETECTED IN OUR EARS?



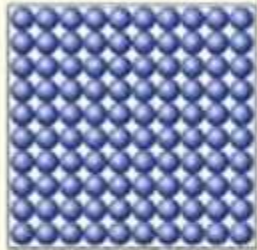
- When a sound wave reaches the ear, a series of high and low pressure regions impinge upon the eardrum.
- The arrival of a compression or high pressure region pushes the eardrum inward;
- The arrival of a low pressure regions serves to pull the eardrum outward.
- The continuous arrival of high and low pressure regions sets the eardrum into vibrational motion.

HOW SOUND WAVES ARE DETECTED IN OUR EARS?

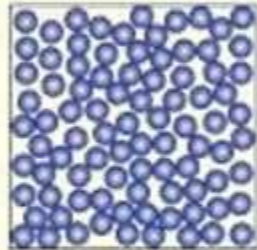


- The eardrum is attached to the bones of the middle ear - the hammer, anvil, and stirrup.
- As these bones begin vibrating, the sound signal is transformed from a pressure wave traveling through air to the mechanical vibrations of the bone structure of the middle ear.
- These vibrations are then transmitted to the fluid of the inner ear
- Where they are converted to electrical nerve impulses which are sent to the brain.

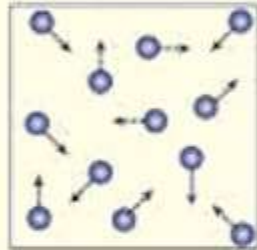
SPEED OF SOUND IN DIFFERENT MEDIUMS



Solid

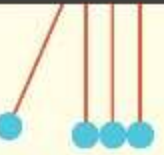


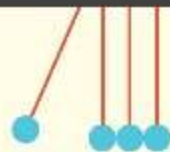
Liquid



Gas

- Two properties that affect wave speed - **inertial properties** and **elastic properties**.
- In general, solids have the strongest interactions between particles, followed by liquids and then gases.
- For this reason, **longitudinal sound waves travel faster in solids** than they do in liquids than they do in gases.
- But, due to inertial property a sound wave will travel **faster in a less dense material** than a more dense material.





QUESTION 2

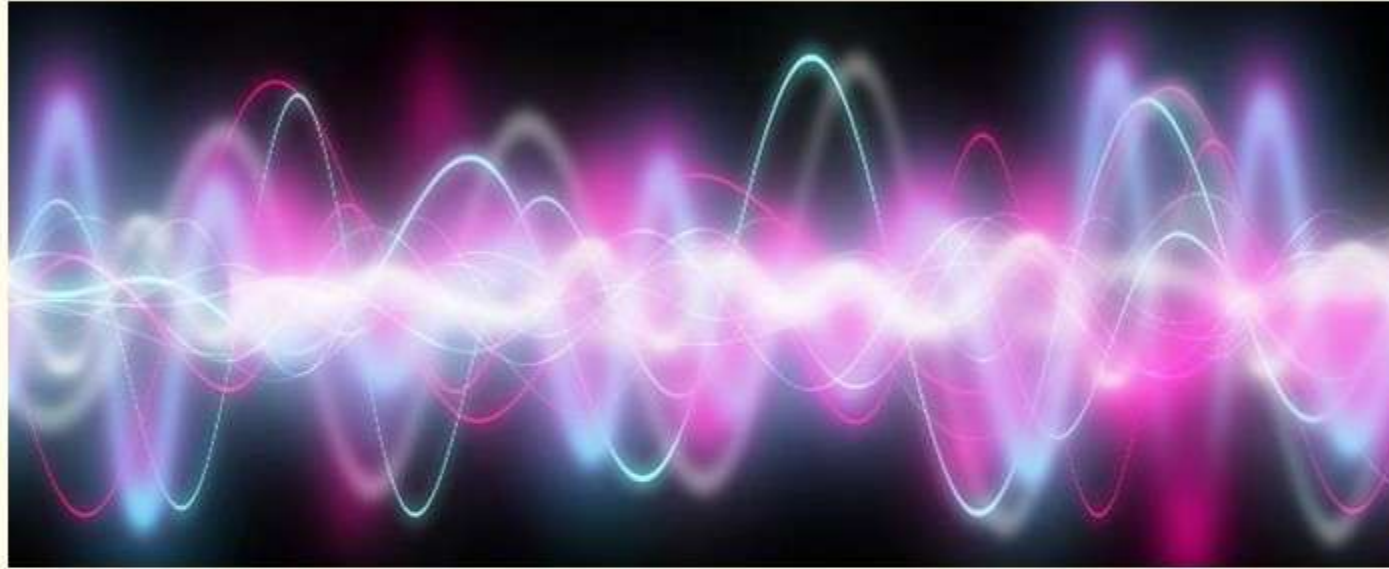
A sound wave will travel nearly three times faster in Helium than it will in air. This is mostly due to:

- a. Helium is a liquid while air is gaseous
- b. Sound waves depend on the elasticity of the material .
- c. Sound can travel faster in noble gases.
- d. Helium particles have lower mass as compared to air particles.

ANSWER : due to inertial property a sound wave will travel faster in a less dense material than a more dense material

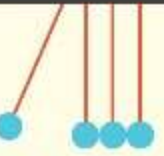


THE WAVE EQUATION



- The mathematical relationship between speed, frequency and wavelength is given by the following equation **Speed = Frequency • Wavelength**
- Using the symbols v , λ , and f , the equation can be rewritten as

$$v = f \cdot \lambda$$



QUESTION 3

If frequency of a sound wave is doubled, what changes

- a. The speed of the sound doubles
- b. The wavelength of the wave doubles.
- c. The speed of the sound halves
- d. The wavelength of the wave halves.

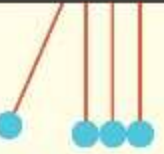
ANSWER: The speed of the sound is constant in a medium, if the frequency changes, it will affect the wavelength but ultimately the speed is unaffected.



HOW DO THESE MUSICAL INSTRUMENTS VIBRATE THE AIR IN IT AND PRODUCE LARGE SOUNDS?



RESONANCE causes a big vibration that is, a loud sound.

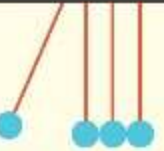


RESONANCE



- Musical instruments are set into vibrational motion at their natural frequency when a person hits, plucks or somehow disturbs the object.
- The natural frequencies of a musical instrument are sometimes referred to as the harmonics of the instrument.
- In Guitars, the vibration of the guitar strings creates standing waves that is passed to the hollow body of the guitar and they both vibrate at the same frequency.
- This is known as resonance - when one object vibrating at the same natural frequency of a second object forces that second object into vibrational motion.

Resonance literally is an induced vibration in an object



RESONANCE HAS STRONG EFFECTS



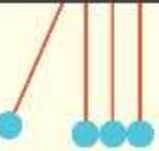
Glasses shatter when they come in contact with sounds that resonates with their natural frequencies.



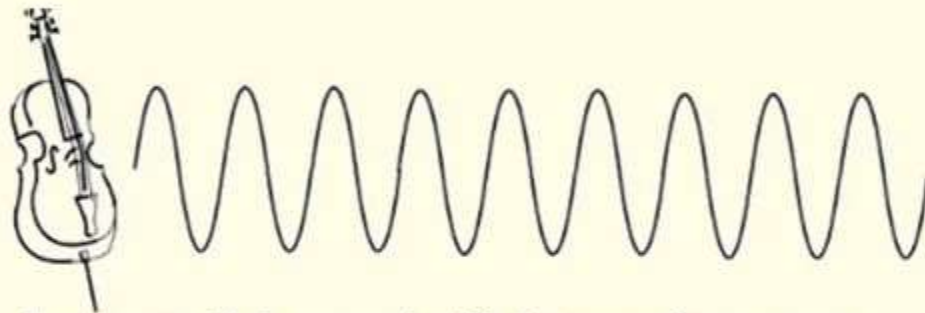
Eardrum can get damaged because of the Resonating sounds of loud speakers



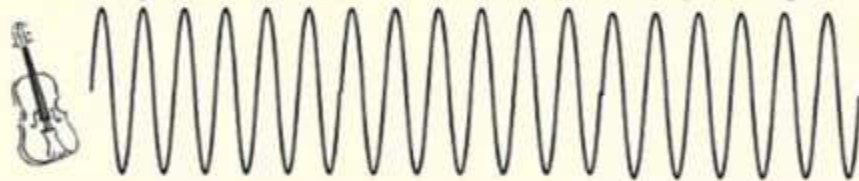
The rhythm of their synchronized footsteps resonates with the natural frequency of the bridge.



WHAT IS THE RELATION BETWEEN PITCH AND FREQUENCY?

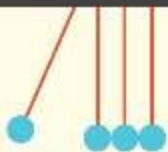
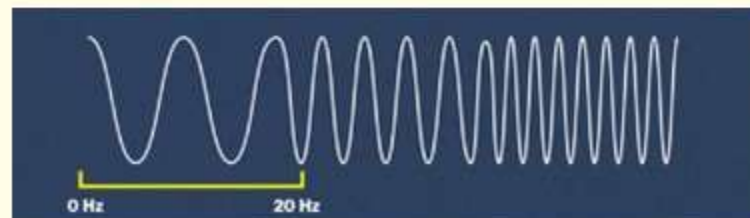


Lower pitch sound with **lower** frequency



Higher pitch sound with **higher** frequency

- Pitch is defined as how our ears can respond to the frequency of a sound wave.
- Pitch increases with frequency.
- Sound waves having a frequency of less than 20Hz is inaudible to human ears as it fail to produce vibrations thus they are called infrasound
- Sound more than 20,000Hz is also inaudible and are called ultrasound.



THE FASCINATING INFRASOUND

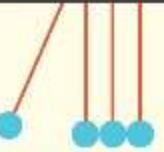


1 Infrasound can be thought of as a slow, periodic rumbling - it is generated by large processes such as Antarctic ice, avalanches, volcanoes, tornadoes, ocean waves, earthquakes, and meteors.

It can be generated by large chemical or nuclear explosions.

2 Numerous animals, including whales, elephants, rhinoceros, giraffes, okapi, and alligators, communicate using infrasound.

3 At the right frequency, infrasound can make human organs vibrate, causing pain. For this reason, it has been a target of weapons research, and various prototypes have been fabricated and tested.



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